

ServiceNow Script-Controlled ACL Project Plan & Phase-Wise Documentation

Project Title: ServiceNow Script-Controlled ACL project for restricting record access based on field values and user conditions.

Team ID: SWTID-2026-2854

Team Size: 5

Team Members

- **Team Leader:** MOHAMED ISTHI A (mohamedisthi19@gmail.com)
 - **Team Member:** Amudhan (amudhansrinivasan02@gmail.com)
 - **Team Member:** DHARUN KUMARP (dd3156684@gmail.com)
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 - **Team Member:** PRIZEGOD V (prizegod2616@gmail.com)
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1. Brainstorming & Ideation Phase

- **Objective:** Identify the core access control challenges in enterprise ServiceNow instances and conceptualize an advanced, script-controlled Access Control List (ACL) solution.
 - **Problem Statement:** Out-of-the-box role-based access controls are often too broad, lacking granular, context-aware restrictions based on dynamic field values (e.g., urgency, category, sensitive data flags) and complex user conditions (e.g., department hierarchy, assignment groups, custom user attributes).
 - **Proposed Solution:** Implement advanced Access Control Lists (ACLs) powered by JavaScript (Server-side scripts using `current` and `gs` objects) to dynamically evaluate read, write, create, and delete permissions.
 - **Key Use Cases:**
 - Restricting incident or case record editing based on priority status and user assignment.
 - Hiding or locking specific sensitive fields unless the user belongs to a specialized security group or manager role.
 - Automating conditional validation prior to record insertion or update.
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2. Requirement Analysis Phase

- **Functional Requirements:**
 - Define target tables (e.g., Incident, Task, or custom tables) for script-controlled ACL application.
 - Implement script-based evaluation logic for `read`, `write`, `create`, and `delete` operations.
 - Integrate user session checks (`gs.getUser()`, `gs.getUserID()`, role memberships, and group assignments).
 - Field-level and table-level security decoupling.
 - **Non-Functional Requirements:**
 - **Performance:** Ensure ACL evaluation scripts are optimized to prevent latency during list and form loads.
 - **Security:** Adhere to ServiceNow best practices (avoiding heavy GlideRecord queries inside ACL scripts where possible, or utilizing caching mechanisms).
 - **Maintainability:** Well-commented server-side scripts with clear error handling and logging.
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3. Project Design Phase

- **Architecture & Components:**
 - **ACL Records (`sys_security_acl`):** Configured for specific operations (`record/read`, `record/write`, `table.*`).
 - **Script Logic:** Utilizing the answer variable pattern (`answer = true;` or `answer = false;`) based on condition evaluation.
 - **Contextual Data Objects:**
 - `current`: Represents the record currently being evaluated.
 - `previous`: Represents the prior state (if applicable).
 - `gs`: GlideSystem API for user session, roles, and logging.
 - **Workflow & Access Matrix:**
 - Table-level validation -> Field-level evaluation -> Final permission grant/deny.
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4. Project Planning Phase

- **Timeline & Task Allocation:**
 - **Phase 1 & 2 (Ideation & Analysis):** Completed by Team Collaboration (All members).
 - **Phase 3 & 4 (Design & Planning):** Led by Mohamed Isthi A and Amudhan.

- **Phase 5 (Development):** Core script development, test record setup, and ACL configuration split among Dharun Kumar P, Premkumar.N, and Prizegod V.
 - **Phase 6 (Testing):** Rigorous simulation with various user personas (ITIL, Admin, End User, Manager) by the entire team.
 - **Phase 7 & 8 (Documentation & Demonstration):** Compilation of GitHub repository structure and video recording preparation.
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5. Project Development Phase

- **Implementation Steps:**

1. **Developer Instance Setup:** Provisioning a ServiceNow Personal Developer Instance (PDI).
2. **Table Configuration:** Creating custom fields or utilizing standard Task/Incident tables with conditional flags (e.g., `u_sensitivity_level`, `u_restricted_access`).
3. **Script-Controlled ACL Implementation:**

- *Sample Table-Level Write ACL Script:*

```
// Check if user is part of the assignment group or has elevated security role
```

```
answer = false;
```

```
if (gs.getUser().hasRole('admin') ||
    current.assignment_group.isMember(gs.getUserID())) {
```

```
    answer = true;
```

```
} else if (current.getValue('u_sensitivity_level') == 'Low') {
```

```
    answer = true;
```

```
}
```

- *Sample Field-Level Read ACL Script:*

```
// Restrict viewing specific financial or confidential fields based on department
```

```
answer = false;
```

```
var userDept = gs.getUser().getDepartment();

if (userDept == 'Finance' || current.opened_by == gs.getUserID()) {

    answer = true;

}
```

4. **Version Control:** Regular code commits and documentation updates pushed to the public GitHub repository.
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6. Project Testing Phase

- **Test Scenarios & Validation:**
 - **Test Case 1:** Regular ITIL user attempting to edit a high-security record -> Expected: Access Denied (**write** ACL evaluates to false).
 - **Test Case 2:** Assigned group member editing the same record -> Expected: Access Granted.
 - **Test Case 3:** Field-level hiding verification for unauthorized departments.
 - **Debugging:** Utilizing **gs.log()** and ServiceNow System Logs to trace ACL evaluation performance and troubleshoot condition failures.
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7. Project Documentation Phase

- **Repository Structure (GitHub Public Repository):**
 - **/ideation** - Brainstorming notes and problem statement outline.
 - **/requirements** - Detailed functional and technical specs.
 - **/design** - Architecture diagrams and access logic workflow charts.
 - **/src** - ACL XML exports, script snippets, and configuration guides.
 - **/testing** - Test case logs and validation results.
 - **README.md** - Comprehensive guide on project setup and execution.
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8. Project Demonstration Phase

- **Demo Video Checklist:**
 - Introduction of Team Members and Team ID (**SWTID-2026-2854**).
 - Statement of Project Name, Purpose, and Enterprise Benefits.

- Screen walkthrough of ServiceNow PDI, ACL configuration screens, and JavaScript logic implementation.
- Live execution testing demonstrating access denial vs. access grant across different user impersonations.
- Final output summary and conclusion.